

ISHMAL KHAN

Mathematics Undergraduate | Data Analyst | Financial Modeling Enthusiast

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PROFESSIONAL SUMMARY

Mathematics graduate of Bahauddin Zakariya University (BZU) with strong expertise in data analysis, financial modeling, and quantitative methods. Proficient in Python, SQL, R, and MATLAB with hands-on experience in machine learning, statistical analysis, time-series forecasting, and financial valuation. Passionate about applying mathematical foundations to real-world data science and finance problems.

EDUCATION

Bachelor of Science in Mathematics

Bahauddin Zakariya University (BZU) | Graduated

Semester GPA: 3.30 | CGPA: 3.17

Relevant Coursework: Calculus, Algebra, Functional Analysis, Discrete Mathematics, Ordinary & Partial Differential Equations, Stochastic Calculus, Complex Analysis, Operations Research, Optimization Theory, Portfolio Theory & Risk Management, Machine Learning, Time Series Analysis, Actuarial Mathematics, Introduction to Deep Learning, Foundations and Applications of Generative AI, Business Administration, Strategic Management, Entrepreneurship

TECHNICAL SKILLS

Programming Languages: Python (Expert), SQL (Advanced), R, MATLAB

Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly

Machine Learning & Stats: Predictive Modeling, Statistical Analysis, Hypothesis Testing, Data Mining

Financial Analysis: Financial Modeling, Portfolio Analysis, Risk Management, DCF Valuation, Time-Series Analysis (ARIMA, Prophet)

Tools & Software: Jupyter Notebook, Microsoft Excel, Power BI, R Studio, AutoCAD, PowerPoint, Canva

FINAL YEAR PROJECT

FraudSense — ML-Based Financial Fraud Detection System

Live App: fraud-frontend-theta.vercel.app

- Built and deployed a full-stack machine learning web application for detecting fraudulent financial transactions in real time
- Supports batch analysis via CSV upload (raw or PCA-transformed data) as well as instant single-transaction fraud checks through a manual entry form
- Trains and evaluates multiple ML models on labeled data, reporting precision, recall, F1-score, and ROC-AUC, with visualizations including ROC curves, confusion matrix, and feature importance
- Applies StandardScaler and PCA preprocessing to raw transaction inputs before generating fraud probability predictions

PROJECTS

Stock Price Forecasting

- Developed time-series forecasting models using ARIMA and Facebook Prophet algorithms
- Created interactive visualizations using Plotly to present forecast results and confidence intervals

Financial Statement Valuation

- Performed comprehensive ratio analysis and Discounted Cash Flow (DCF) valuation
- Implemented financial models in Microsoft Excel and Python for real company data

MCB Historical Prices — Exploratory Analysis

- Conducted exploratory data analysis on MCB Bank historical stock price data
- Identified long-term trends, seasonality patterns, and notable price movements

Stock Price Simulation (MCB) — Monte Carlo

- Applied Monte Carlo simulation methods to model possible future stock price paths for MCB Bank
- Quantified risk and uncertainty across thousands of simulated price trajectories

MCB Return Distribution Analysis

- Computed and visualized daily log returns for MCB Bank stock
- Produced statistical summary including distribution shape, skewness, and volatility metrics

Interactive Dashboards

- Built exploratory data analysis dashboards using Plotly Dash and Power BI
- Designed interactive visualizations enabling dynamic data exploration

ACHIEVEMENTS

BusinessEXPO Event Organizer

- Organized and coordinated the BusinessEXPO event, demonstrating leadership, planning, and execution skills in a competitive business-focused environment

Best Girls Representative Award

- Recognized as Best Girls Representative for outstanding dedication and advocacy within the student community

LANGUAGES

English (Proficient)